PDE BASED IMAGE RECONSTRUCTION

James Rosado

Numerical Partial Differential Equations
Spring 2020



Table of Contents

1. Introduction
2. Analytic Formulation
3. Gaussian Smoothing
4. Numerical Solving
5. Results

Project Goals

- Implement an active contour method to delineate an image from noise.
- Determine necessary parameters to improve the detection of image contours and investigate the robustness of the active contour method.

Project Steps

- Read in a raw image and process the image through Gaussian Smoothing with variance σ [13].
- Set an edge detector function $g(\cdot)$ where $\lim_{|\vec{z}|\to\infty} g(\vec{z}) = 0$ and choose parameters [13]
- Apply a level set method to move an active contour (a front, snake) towards the image using the image curvature and edge detector.

Active Contour Model

- Raw image $u_0(x)$, where $x \in \Omega \subset \mathbb{R}^2$ in 2D
- Active contour C minimizes:

$$F(C) = \underbrace{\alpha \int_0^1 |C'(s)|^2 \, ds + \beta \int_0^1 |C''(s)| \, ds}_{\text{control smoothness}} - \underbrace{\lambda \int_0^1 |\nabla u_0(C(s))|^2 \, ds}_{\text{controls movement of } C} \tag{1}$$

- C realizes the image
- Move C using edge detector $g(\cdot)$ and curvature κ
- Edge Detector [13], two parameters γ and p

$$g(\nabla u_0(\vec{x})) = \frac{1}{1 + \gamma |J_\sigma \star \nabla u_0|^p} \tag{2}$$

- J_σ is the Gaussian Smoothing, but why?

Active Contour Model

- Equation (1) is equivalent to [13, 16]

$$\phi_t = |\nabla\phi|\nabla \cdot \underbrace{\left[g(\nabla u_0)\left(\frac{\nabla\phi}{|\nabla\phi|}\right)\right]}_{\text{generalized curvature}}$$

$$= |\nabla\phi|\left(g(\nabla u_0)\nabla \cdot \frac{\nabla\phi}{|\nabla\phi|} + \nabla g(\nabla u_0)\cdot\frac{\nabla\phi}{|\nabla\phi|}\right) \tag{3}$$

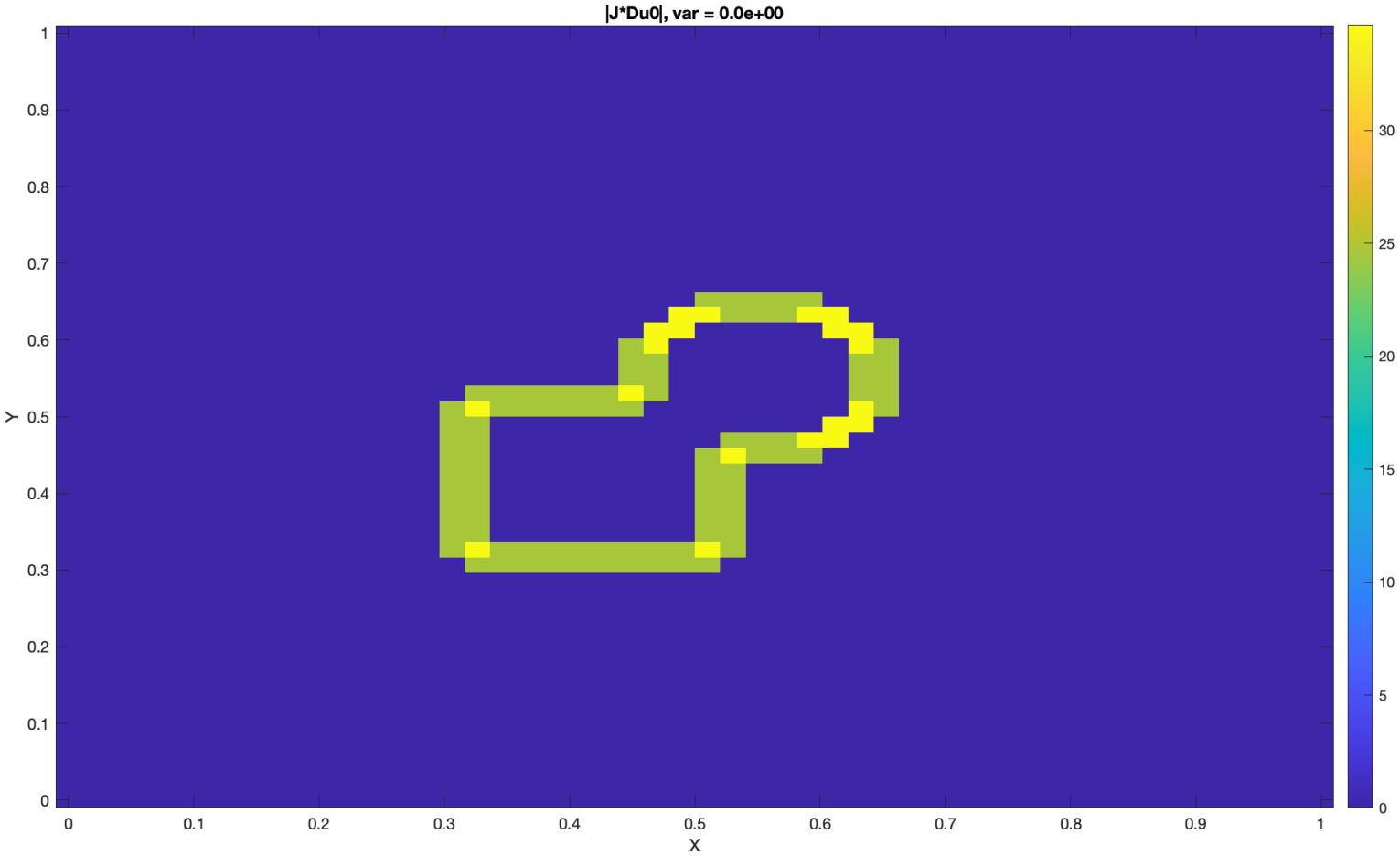
$$= |\nabla\phi|\left(g(\nabla u_0)\kappa + \nabla g(\nabla u_0)\cdot\frac{\nabla\phi}{|\nabla\phi|}\right)$$

$$\phi_t = |\nabla\phi| g(\nabla u_0)\kappa + \nabla g(\nabla u_0)\cdot\nabla\phi \tag{4}$$

- ϕ is the level set function
- $\Gamma(t)$ is the contour corresponding to $\phi(x,t) = 0$, which will outline our image.
- Use (3) to write numerical approximation

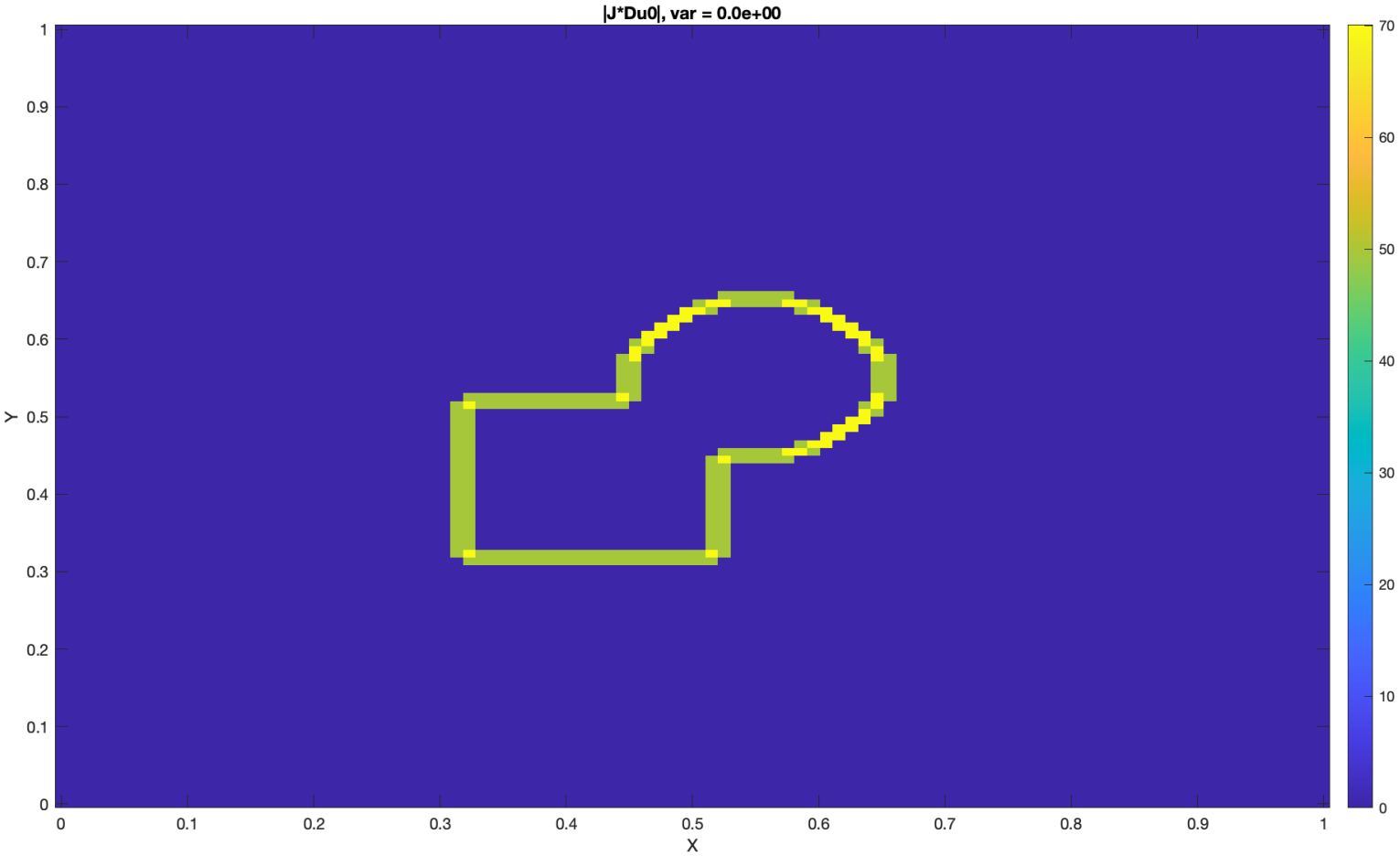
Smoothing An Image

- ∇u_0 has order $\mathcal{O}(1/h)$ grows unbounded as $h \to 0$



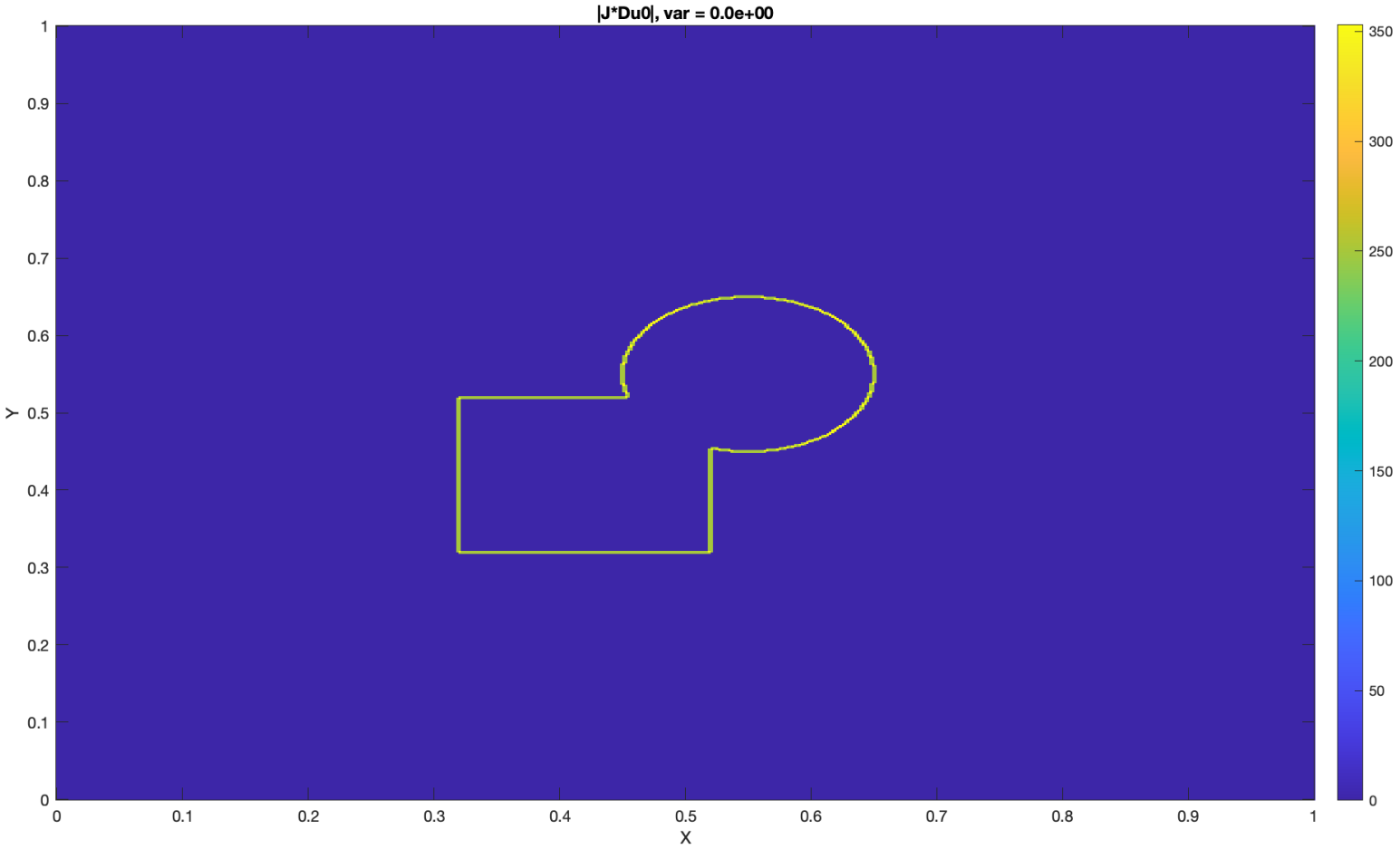
Smoothing An Image

- ∇u_0 has order $\mathcal{O}(1/h)$ grows unbounded as $h \to 0$



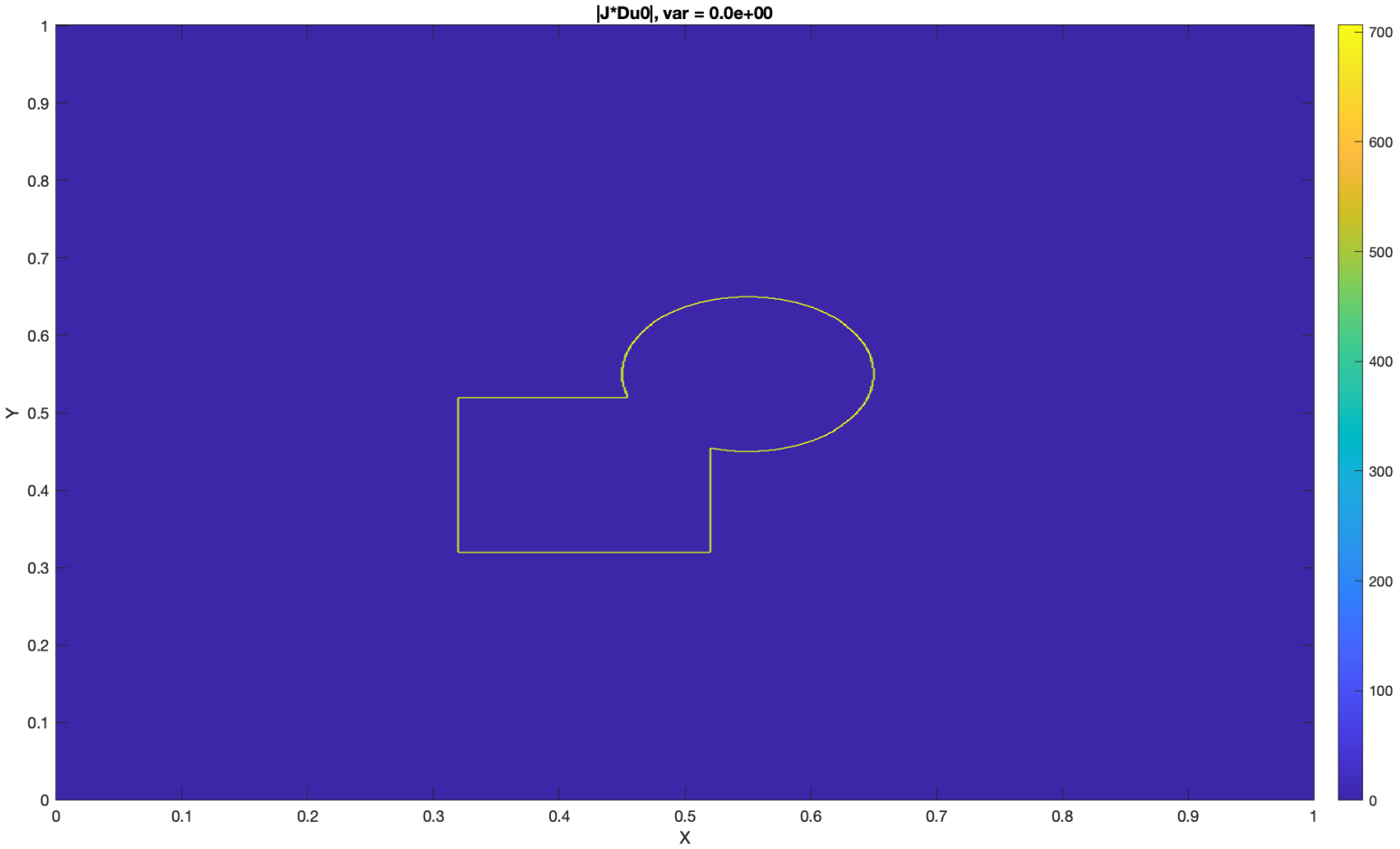
Smoothing An Image

- ∇u_0 has order $\mathcal{O}(1/h)$ grows unbounded as $h \to 0$



Smoothing An Image

- ∇u_0 has order $\mathcal{O}(1/h)$ grows unbounded as $h \to 0$



Gaussian Smoothing

- Numerically solve heat equation up to time $t = \sigma$, $u_t = -\nabla^2 u$
- Crank-Nicolson, two dimensions analogous to LeVeque [10]
- $h_x = h_y = h$:

$$\frac{U_{i,j}^{n+1} - U_{i,j}^{n}}{k} = \frac{1}{2}\left(\nabla^2 U_{i,j}^{n} + \nabla^2 U_{i,j}^{n+1}\right)$$

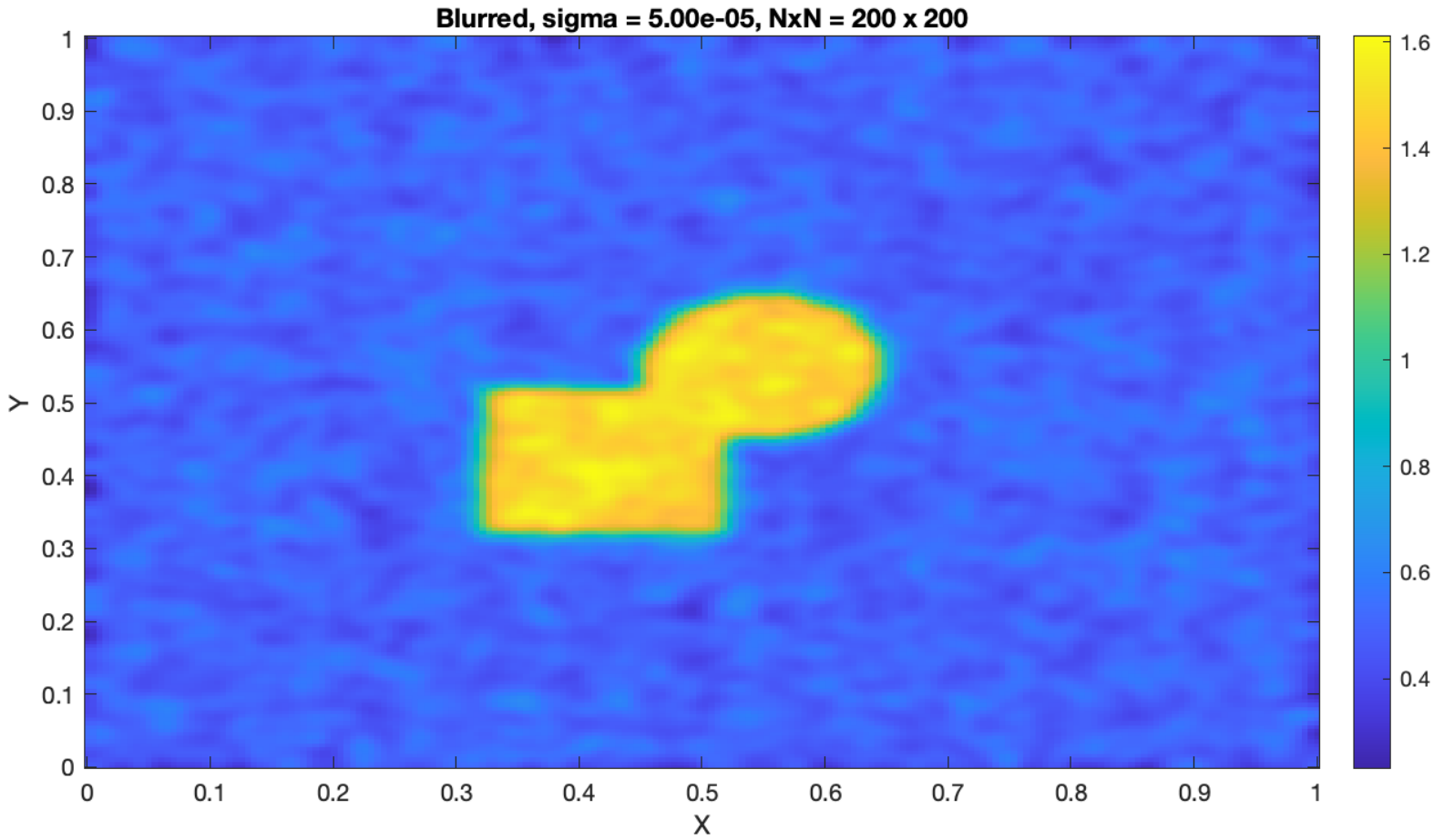
where

$$\nabla^2 U_{i,j}^{n} = \frac{U_{i-1,j}^{n} - 2U_{i,j}^{n} + U_{i+1,j}^{n}}{h^2} + \frac{U_{i,j-1}^{n} - 2U_{i,j}^{n} + U_{i,j+1}^{n}}{h^2}$$

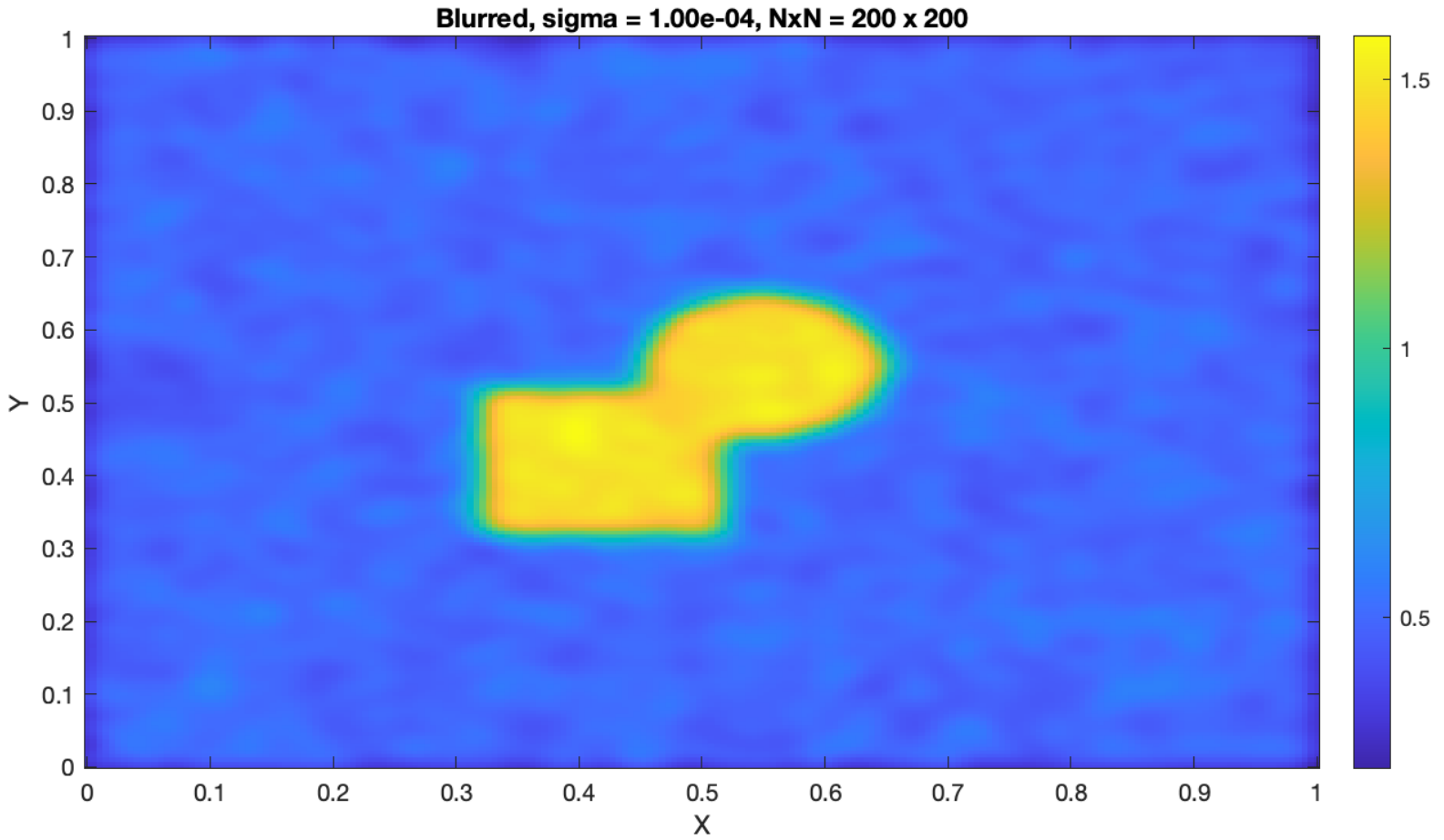
$$\nabla^2 U_{i,j}^{n+1} = \frac{U_{i-1,j}^{n+1} - 2U_{i,j}^{n+1} + U_{i+1,j}^{n+1}}{h^2} + \frac{U_{i,j-1}^{n+1} - 2U_{i,j}^{n+1} + U_{i,j+1}^{n+1}}{h^2}$$

- Constant Neumann boundary conditions

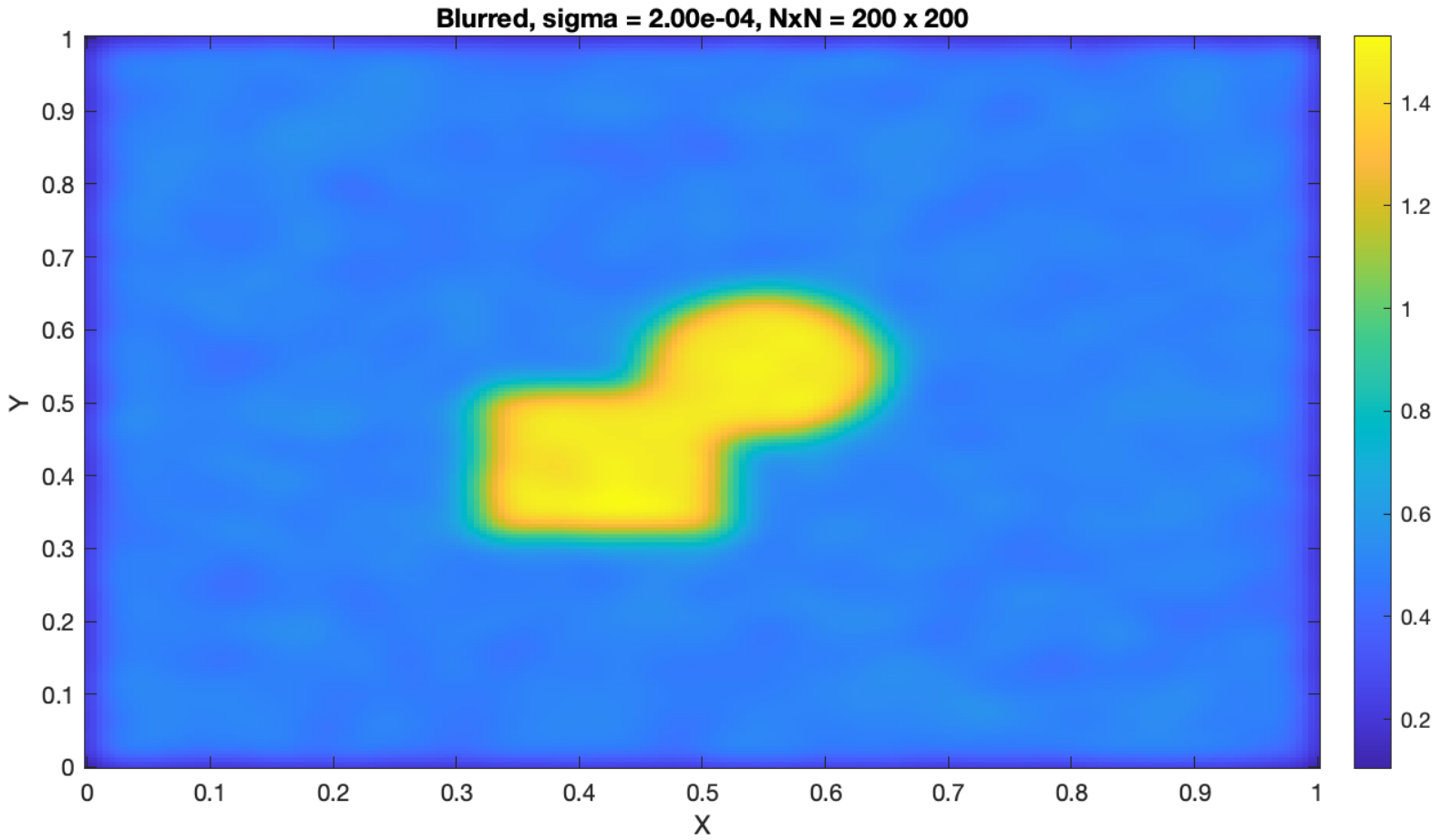
Gaussian Smoothing



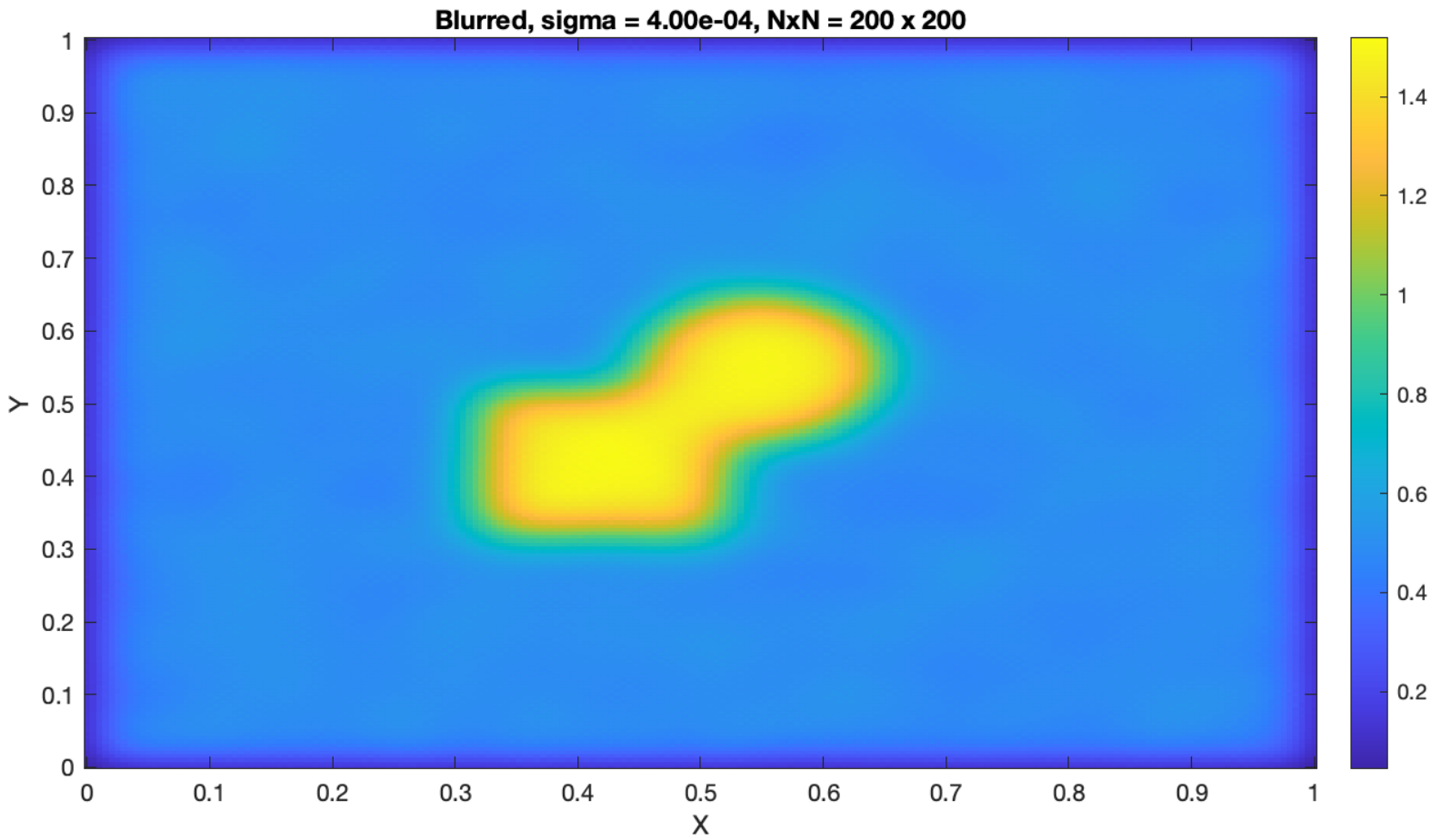
Gaussian Smoothing



Gaussian Smoothing



Gaussian Smoothing



Gaussian Smoothing

- How are variance σ and smoothing width d related?
- analytically solve heat equation, Dirac initial data:

$$\begin{cases} u_t = -\nabla^2 u & \vec{x} \in \Omega, t > 0 \\ u(x,0) = \delta_0(x) \end{cases}$$

- Fundamental solution [5]:

$$\Phi(\vec{x}, t) = \frac{1}{4\pi t} e^{-|\vec{x}|^2/(4t)}$$

- Amount of smoothing is determined by $t = \sigma > 0$; therefore,

$$\Phi(\vec{x}, \sigma) = \frac{1}{4\pi\sigma} e^{-|\vec{x}|^2/(4\sigma)}$$

Gaussian Smoothing

- Maximum when $\vec{x} = 0$; therefore, $\Phi_{max} = 1/(4\pi\sigma)$.
- Width of the smoothing can be given by when Φ is 5% of the maximum
- solve

$$0.05 \cdot \frac{1}{4\pi\sigma} = \frac{1}{4\pi\sigma} e^{-d^2/(4\sigma)}$$

  which has solution given by $d(\sigma) = 2\sqrt{\sigma \ln(20)}$
- Grid resolution $h < d$ to resolve smoothing region.
- Choose p and γ such that $|\nabla g(\nabla u_0)|$ is order magnitude of $1/d$.

Numerical Scheme

- Write out the terms for the *generalized curvature* (3):

$$\underbrace{A\nabla \cdot \frac{\nabla\phi}{|\nabla\phi|}}_{\text{Part 1}} + \underbrace{\nabla A \cdot \frac{\nabla\phi}{|\nabla\phi|}}_{\text{Part 2}}$$

  where $A = g(\nabla u_0)$

- Part 1

$$= A\left(\partial_x\left(\frac{\phi_x}{(\phi_x^2+\phi_y^2)^{1/2}}\right) + \partial_y\left(\frac{\phi_y}{(\phi_x^2+\phi_y^2)^{1/2}}\right)\right)$$

$$= A\left(\frac{|\nabla\phi|\phi_{xx} - \phi_x\frac{1}{|\nabla\phi|}(\phi_x\phi_{xx}+\phi_y\phi_{yx})}{|\nabla\phi|^2} + \cdots\right)$$

$$= \frac{A}{|\nabla\phi|^3}\left(|\nabla\phi|^2\phi_{xx} - \phi_x^2\phi_{xx} - 2\phi_x\phi_y\phi_{xy} - \phi_y^2\phi_{yy} + |\nabla\phi|^2\phi_{yy}\right)$$

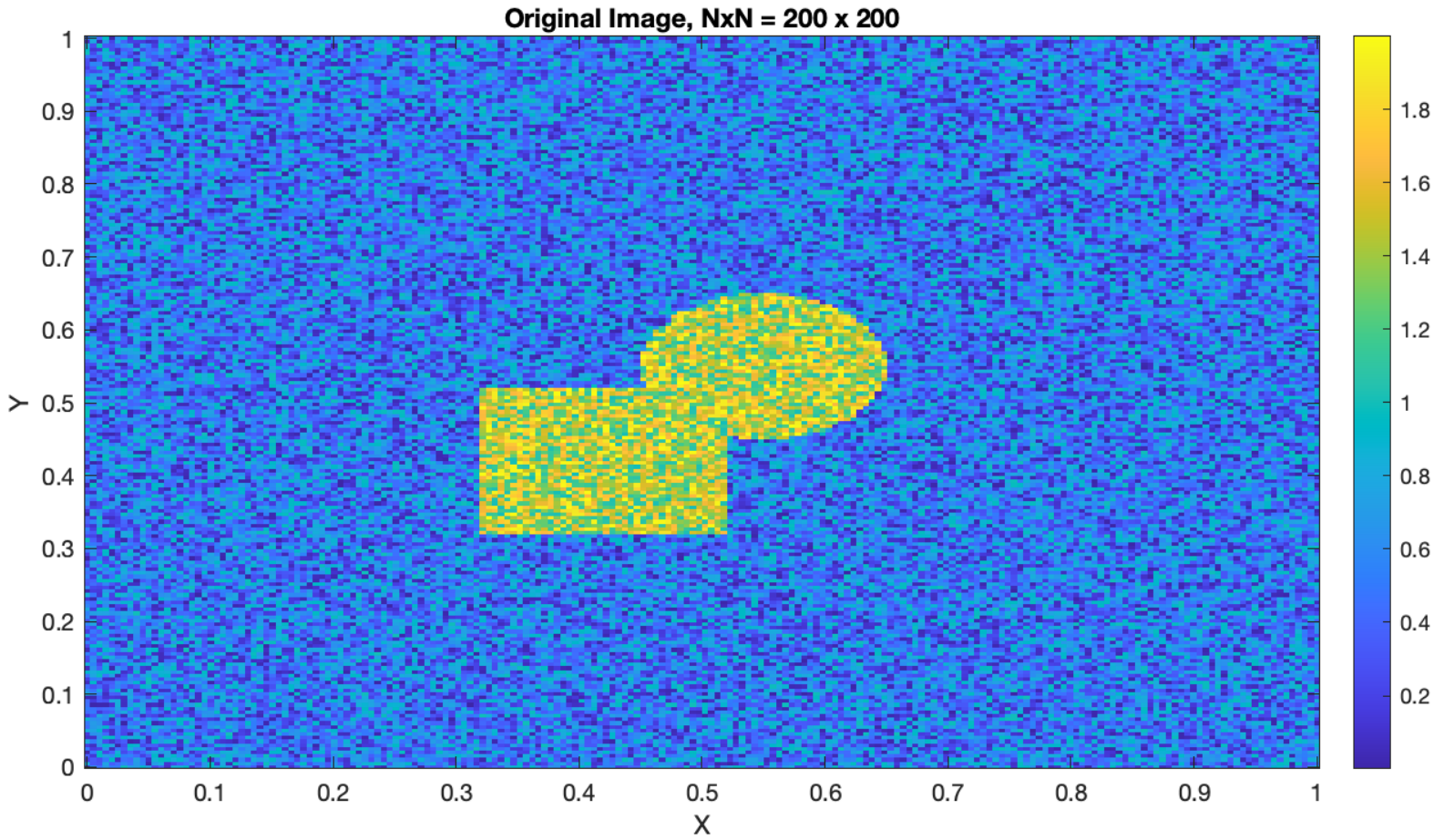
Numerical Scheme

- Use central finite differences to approximate first and second derivatives
- Part 2 use center differences for the gradient of A

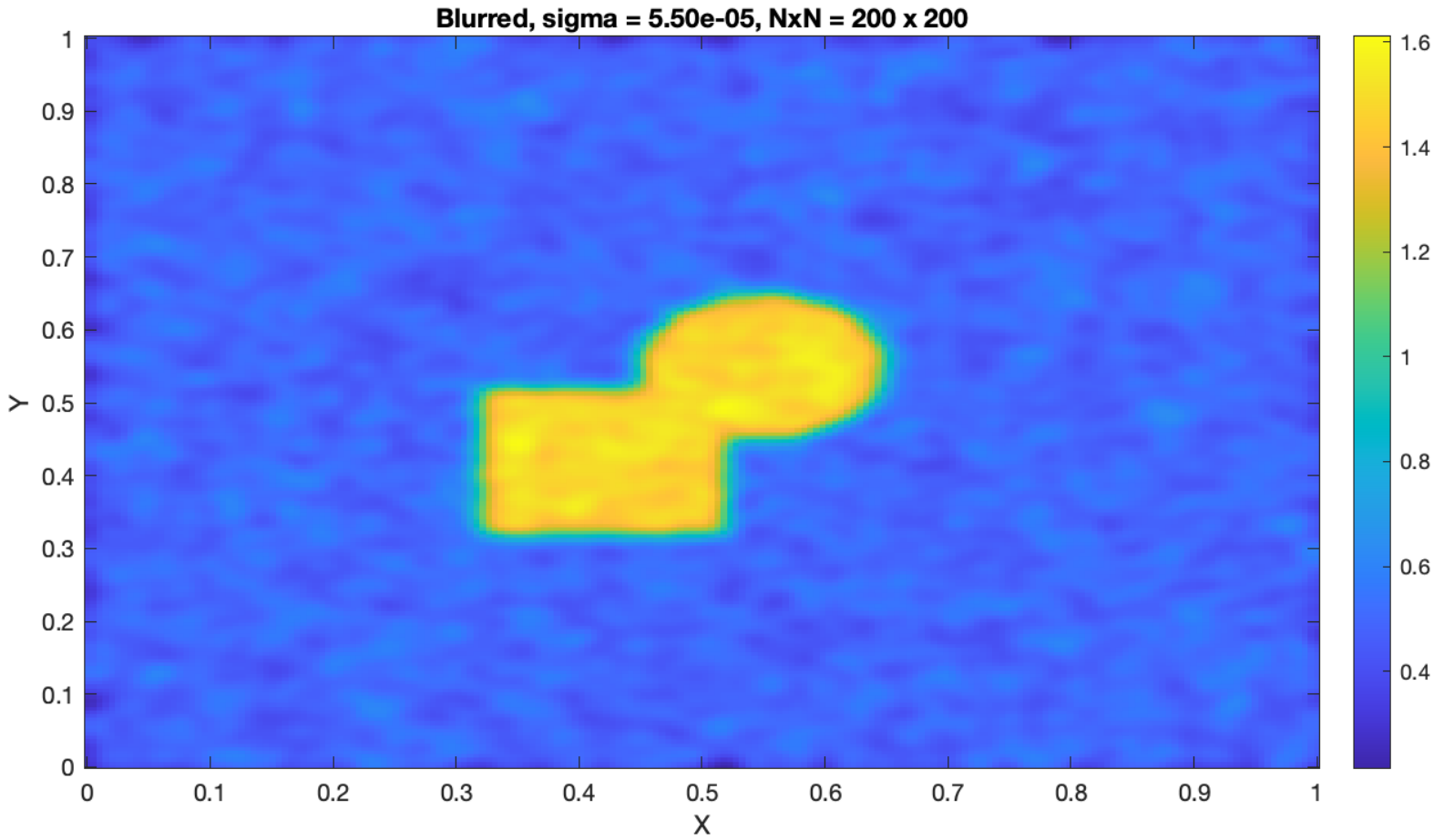
$$(A_x, A_y) \cdot \left(\frac{\phi_x}{\sqrt{\phi_x^2 + \phi_y^2}^{1/2}}, \frac{\phi_y}{\sqrt{\phi_x^2 + \phi_y^2}^{1/2}} \right)$$

- Hamilton Jacobi Scheme in 2D
- Level set update is FE.

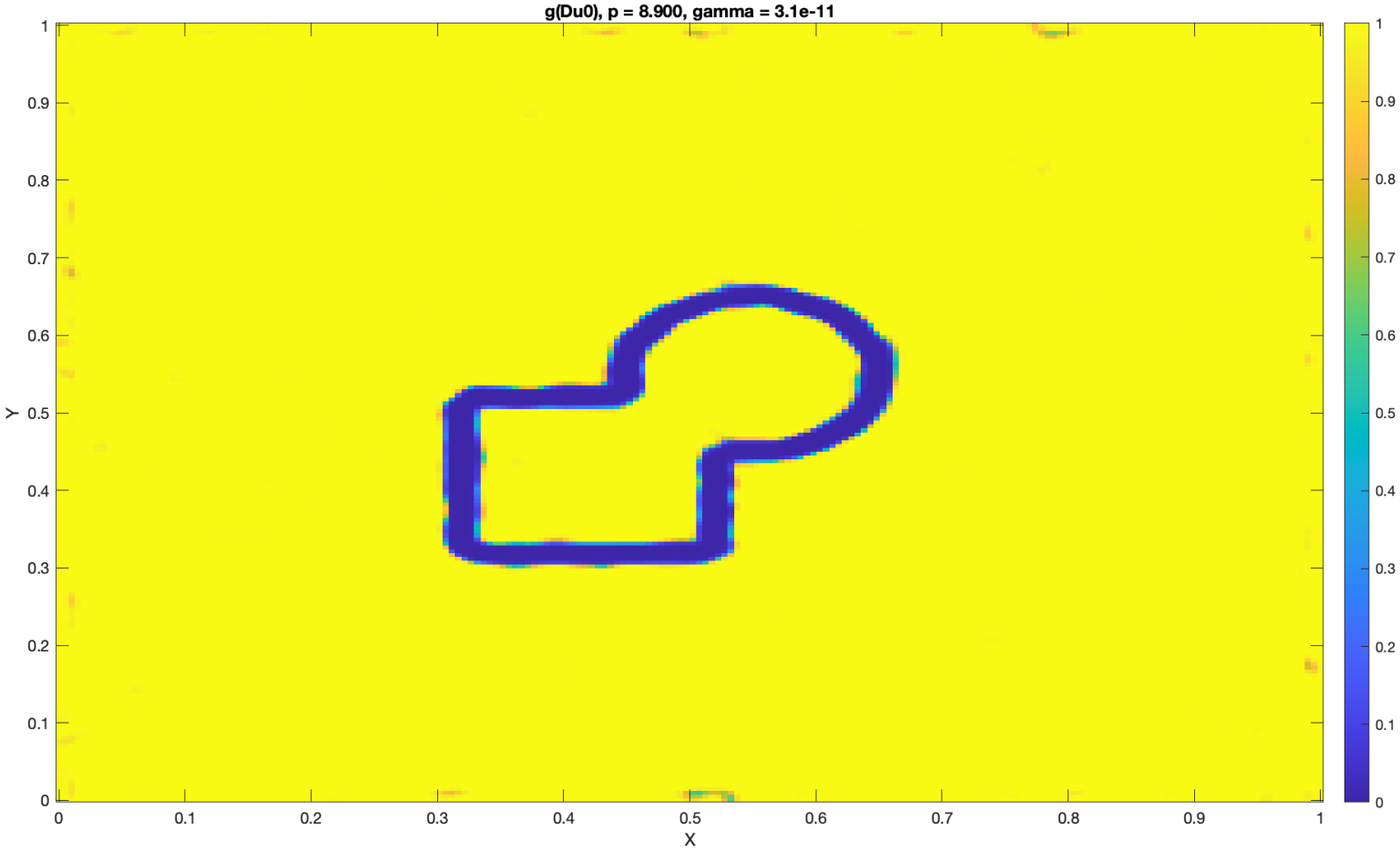
Run #1: Some Noise



Run #1: Some Noise



Run #1: Some Noise

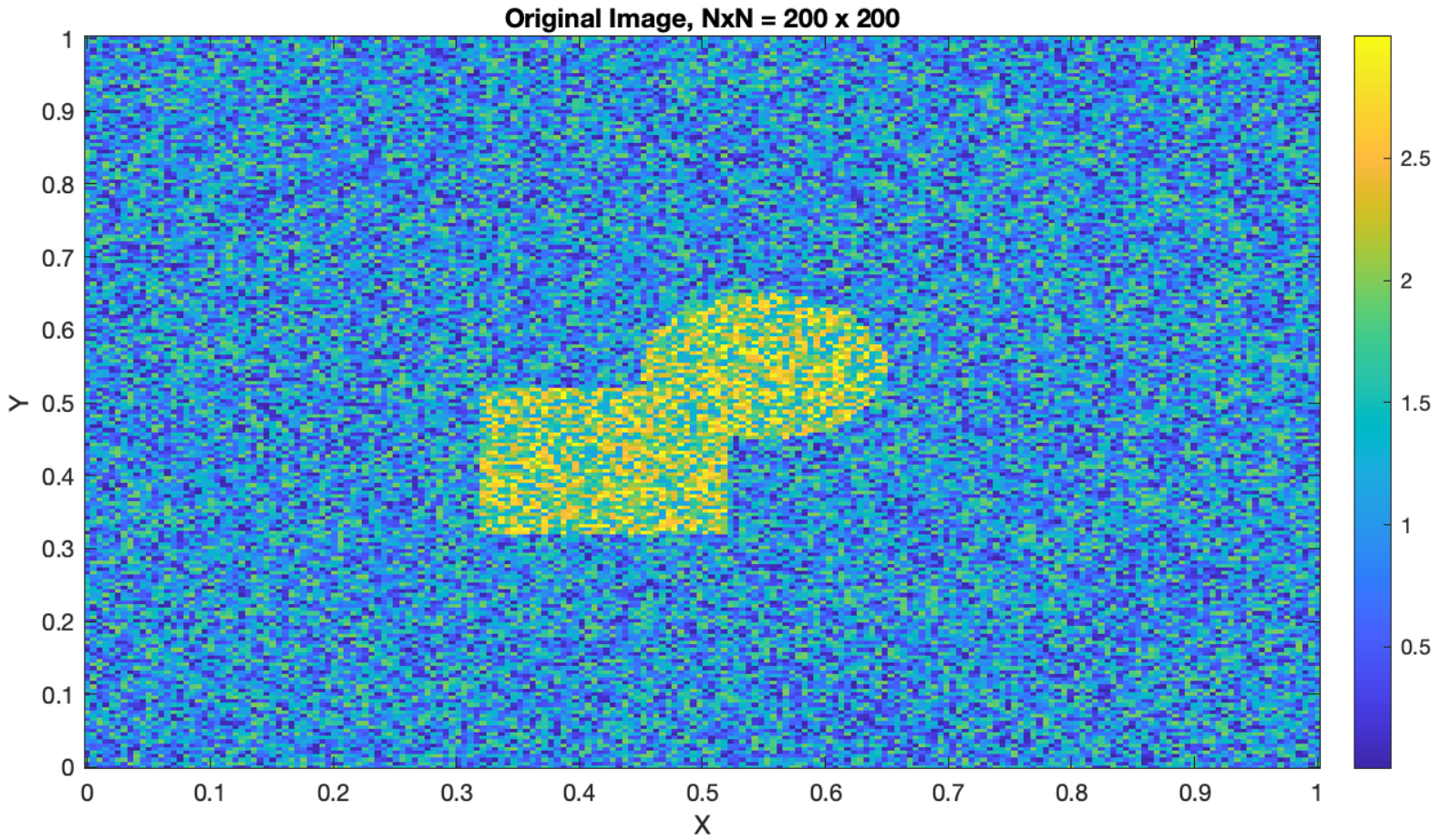


Run # 1: Some Noise

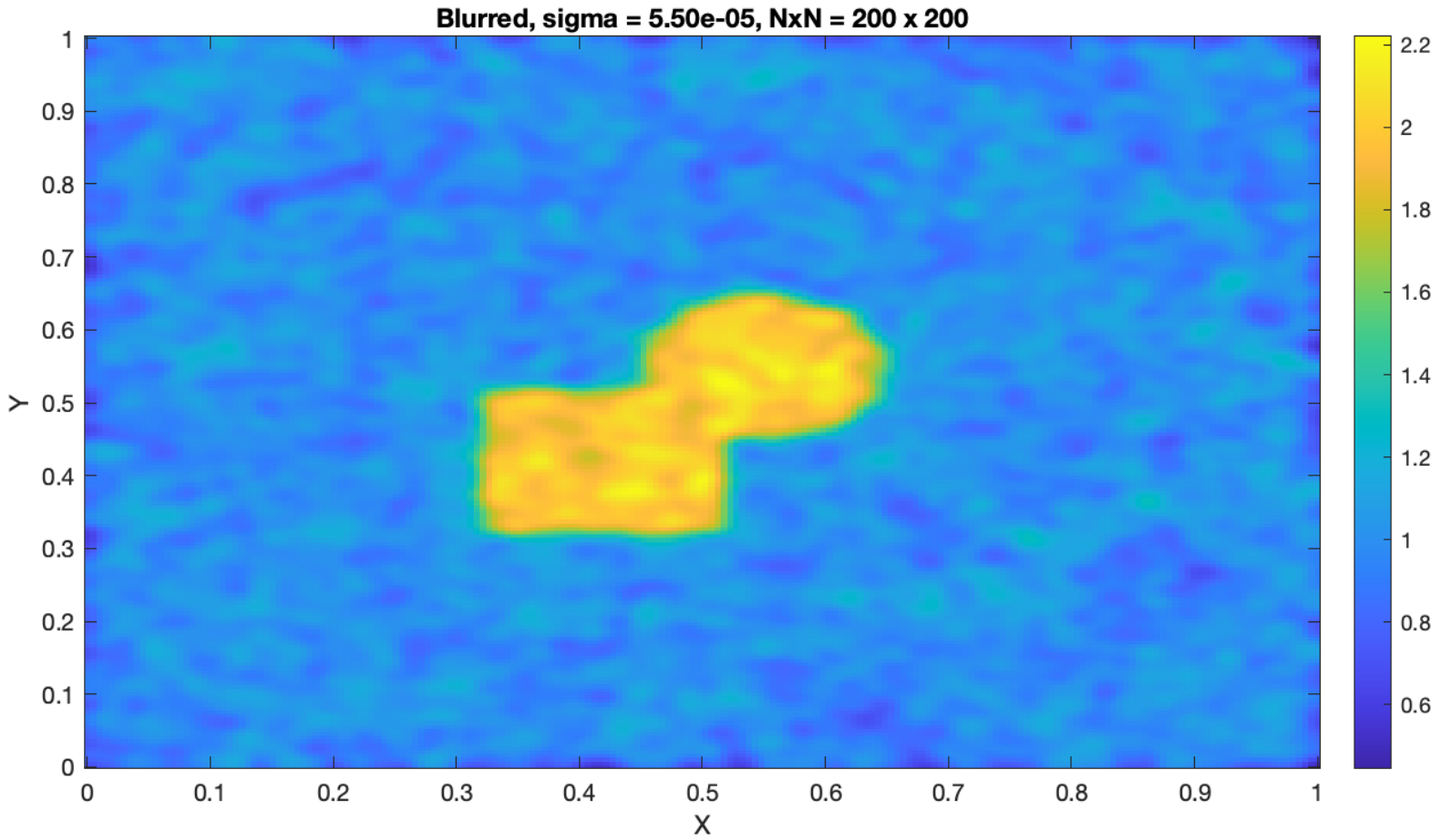


- $\sigma = 5.5 \times 10^{-5}$, $p = 8.9$, $\gamma = 5.5 \times 10^{-11.25}$

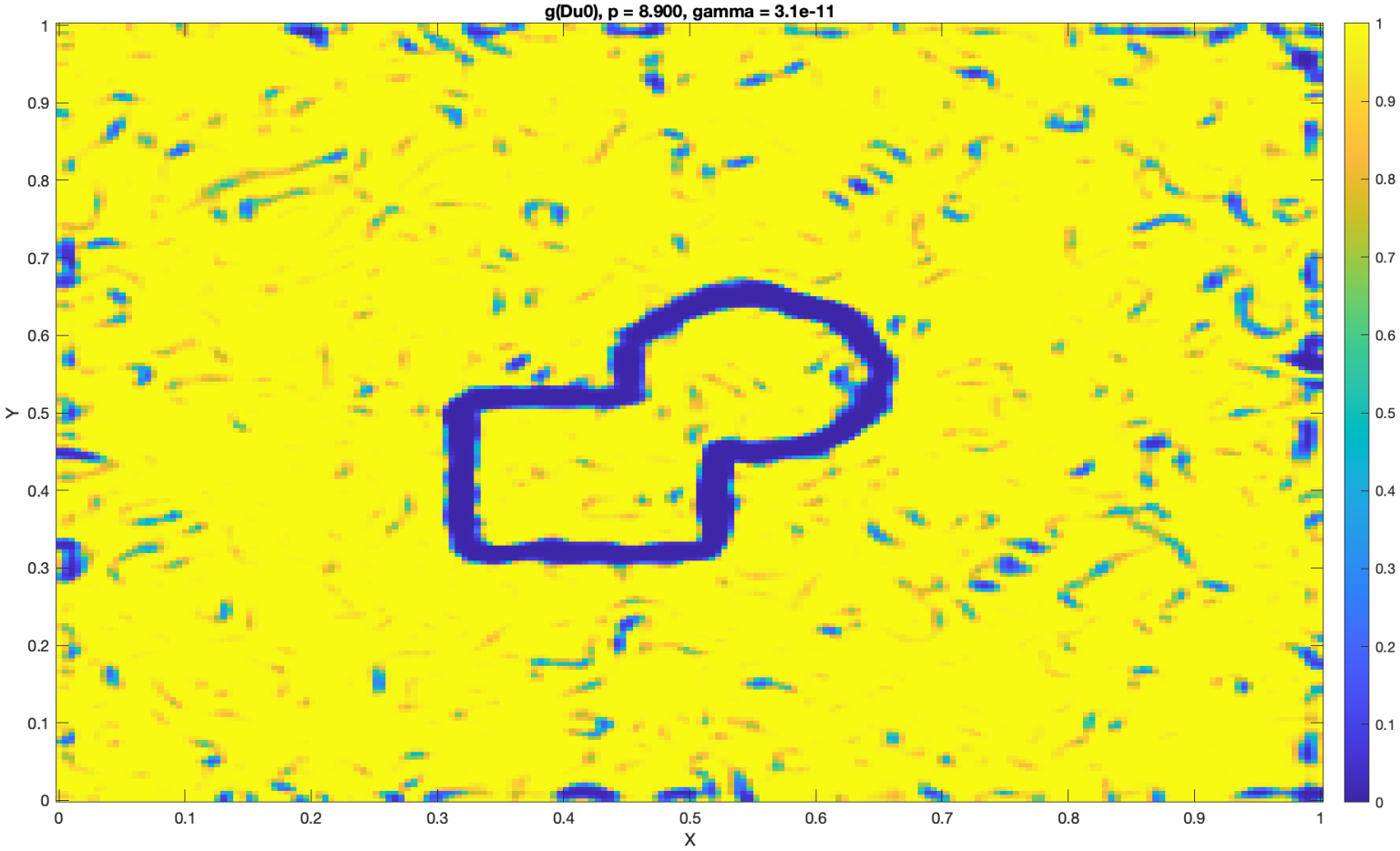
Run #2: More Noise, same parameters



Run #2: More Noise, same parameters



Run #2: More Noise, same parameters

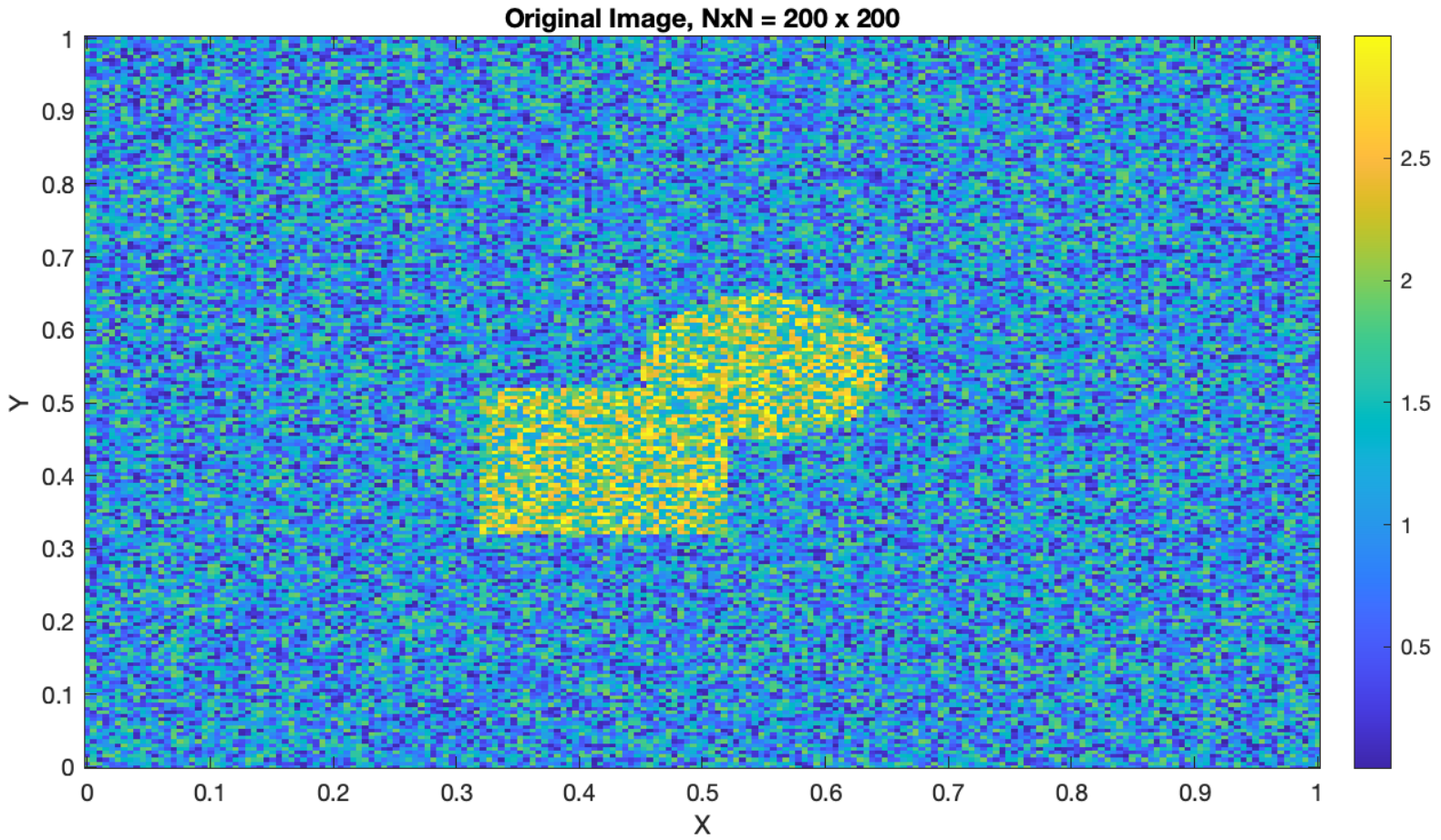


Run #2: More Noise, same parameters

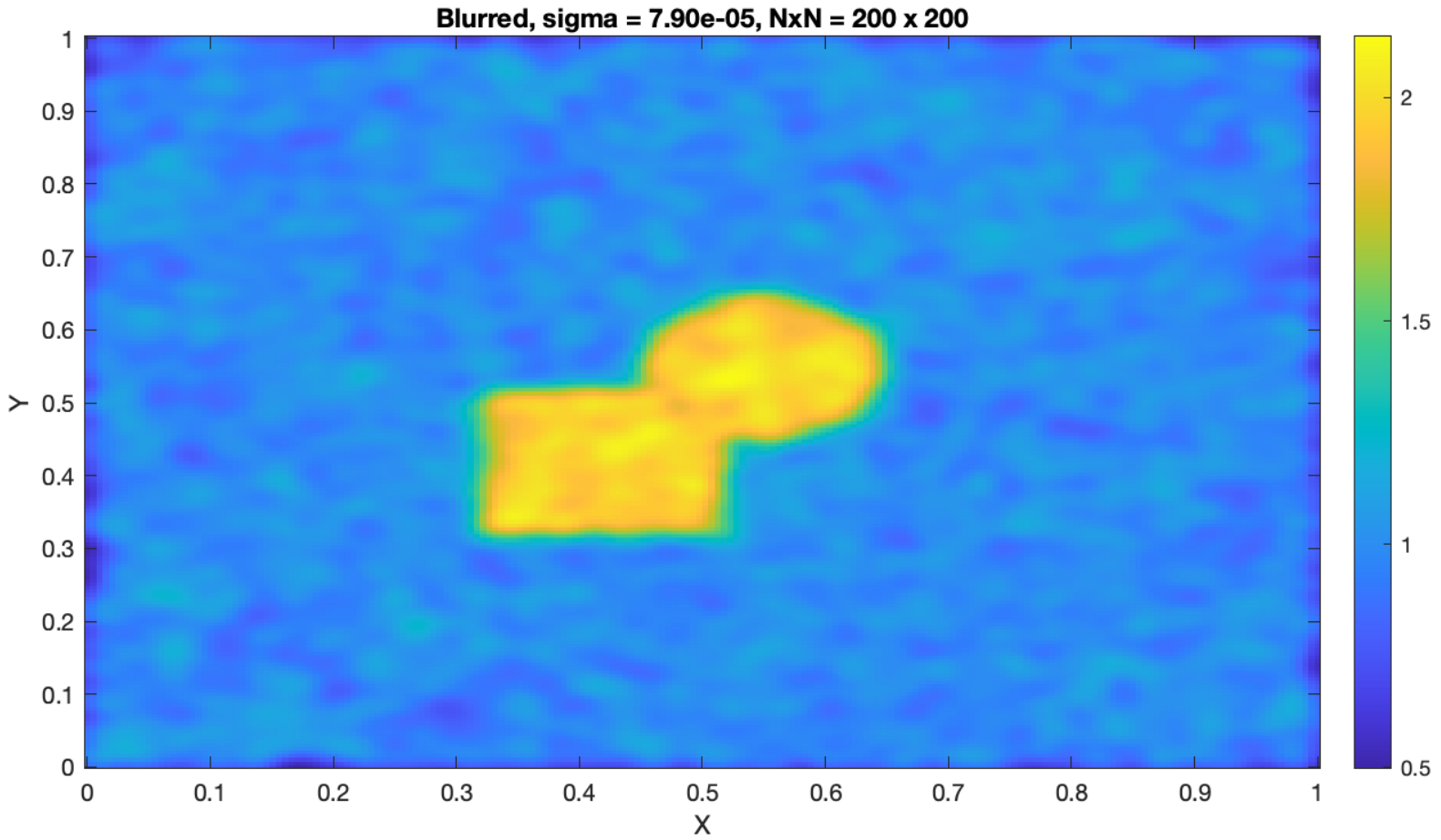


- $\sigma = 5.5 \times 10^{-5}$, $p = 8.9$, $\gamma = 5.5 \times 10^{-11.25}$

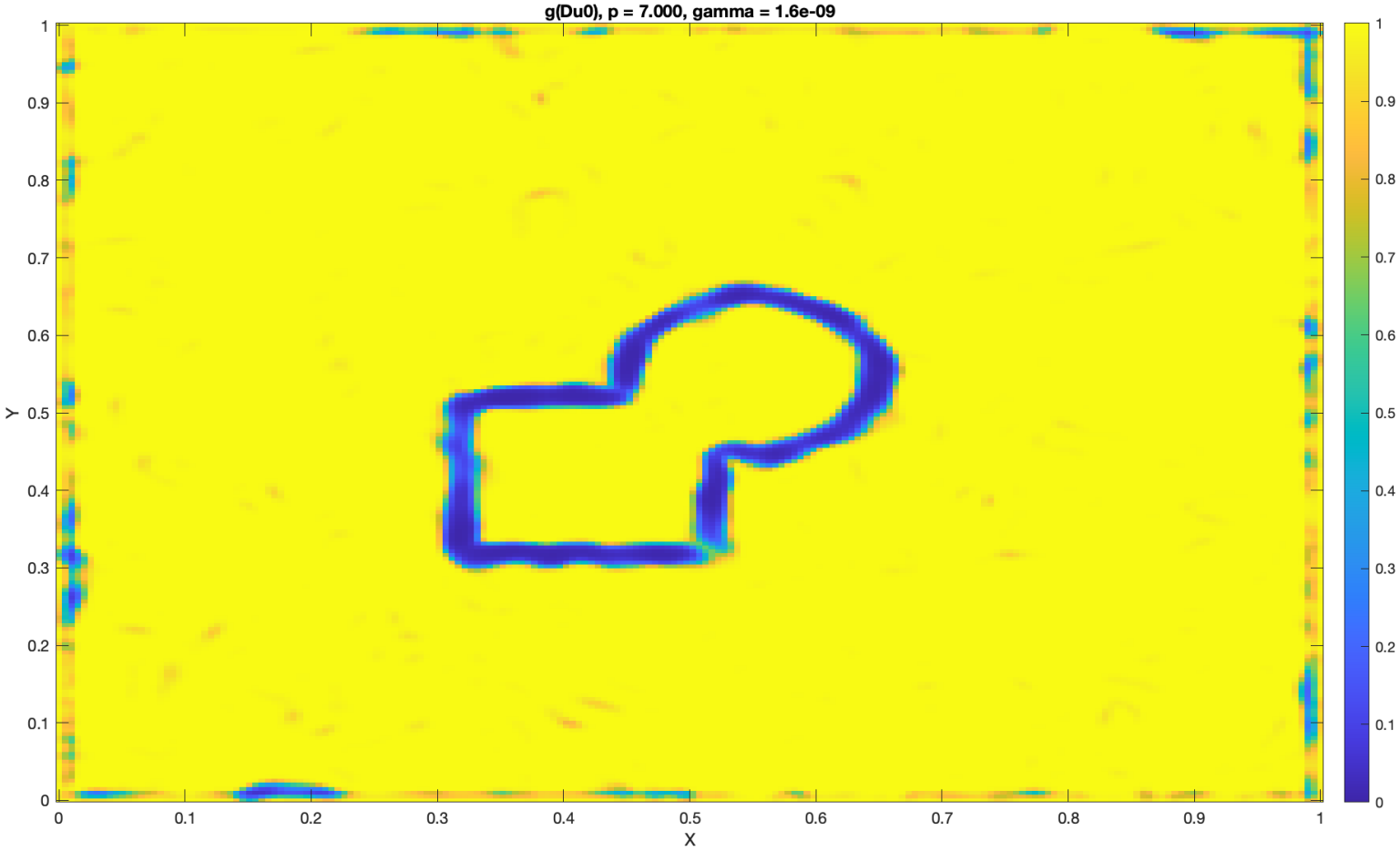
Run #3: More Noise, modified parameters



Run #3: More Noise, modified parameters



Run #3: More Noise, modified parameters

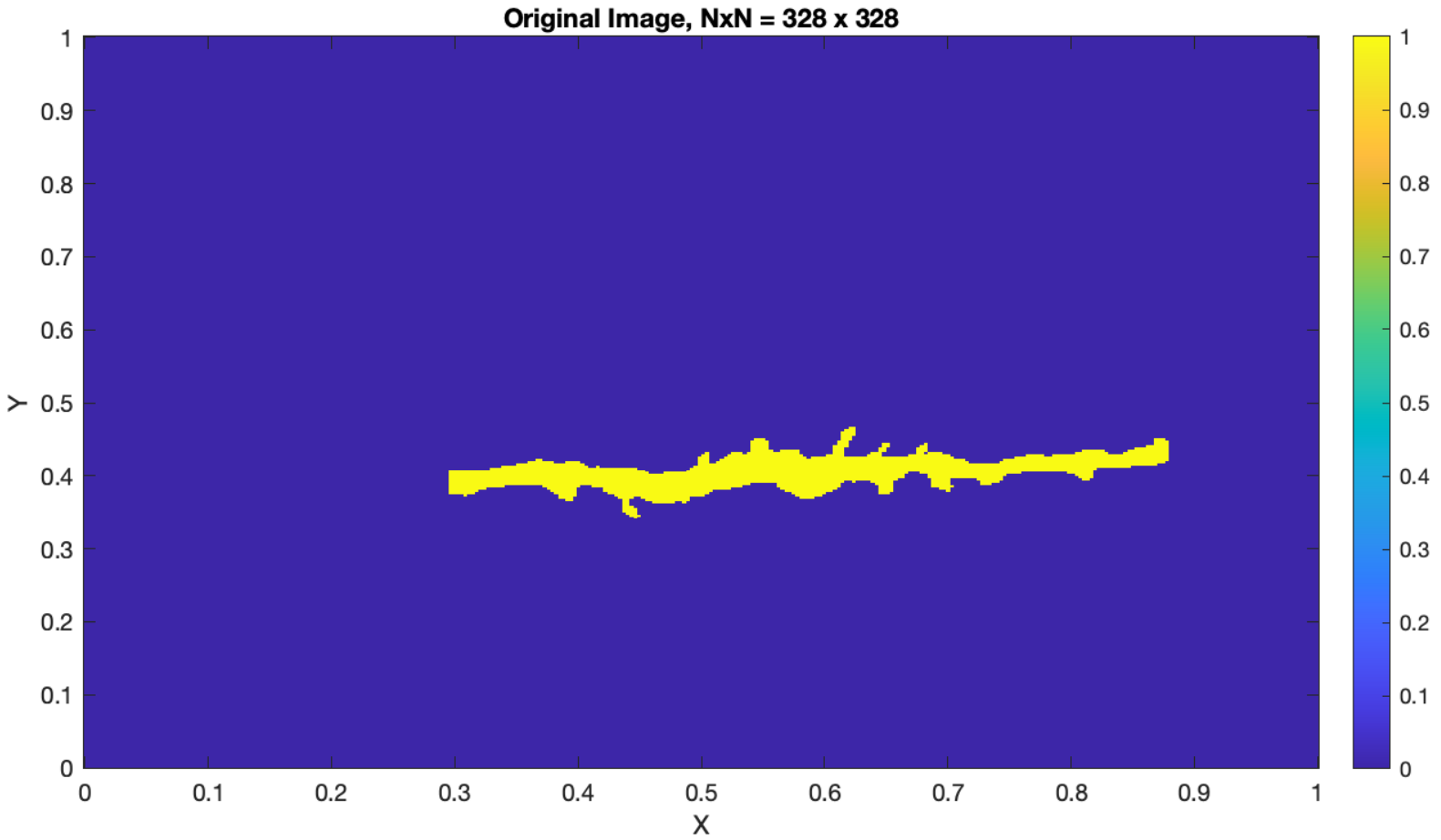


Run #3: More Noise, modified parameters

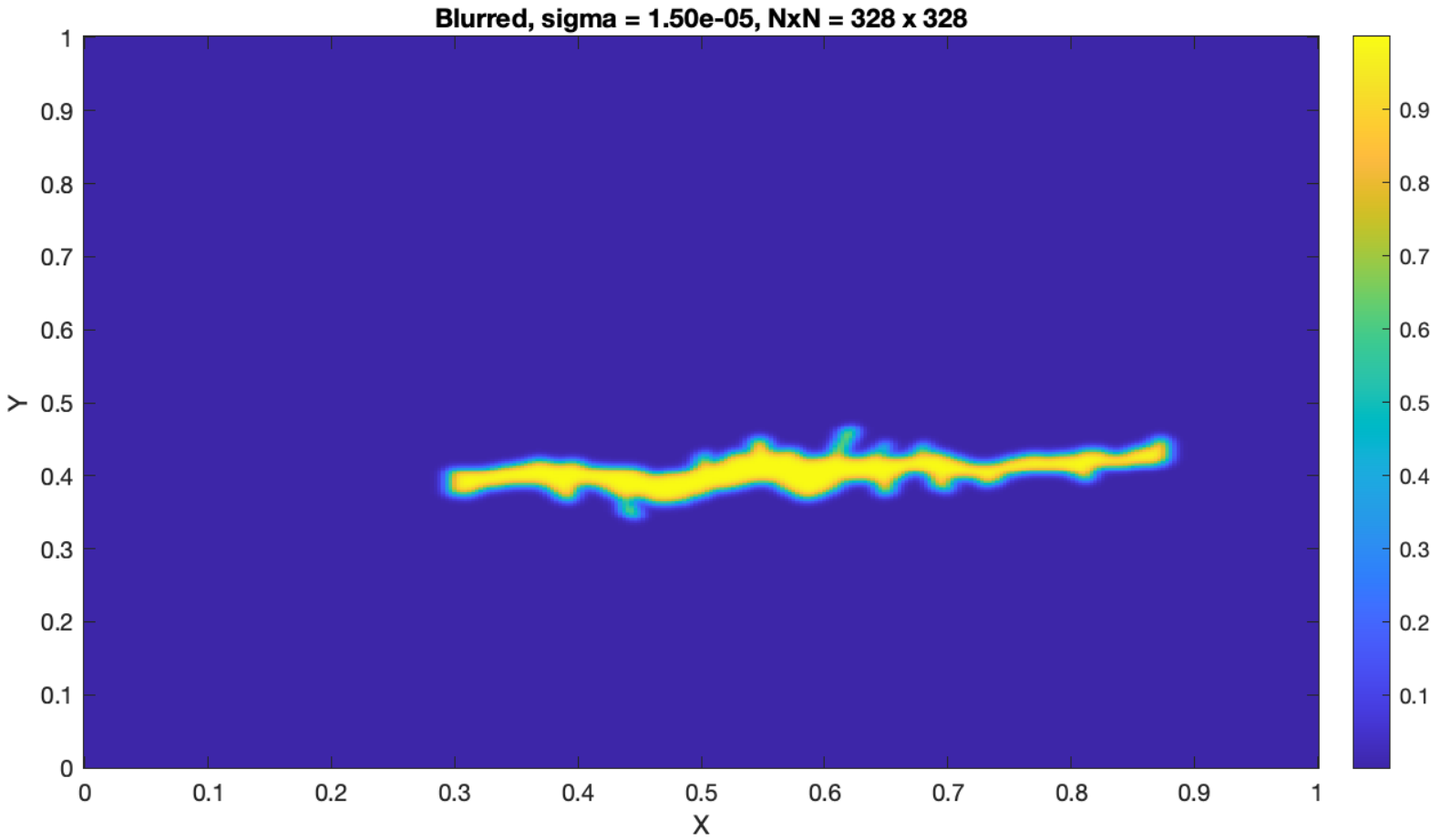


- $\sigma = 7.9 \times 10^{-5}$, $p = 7.0$, $\gamma = 5.5 \times 10^{-9.55}$

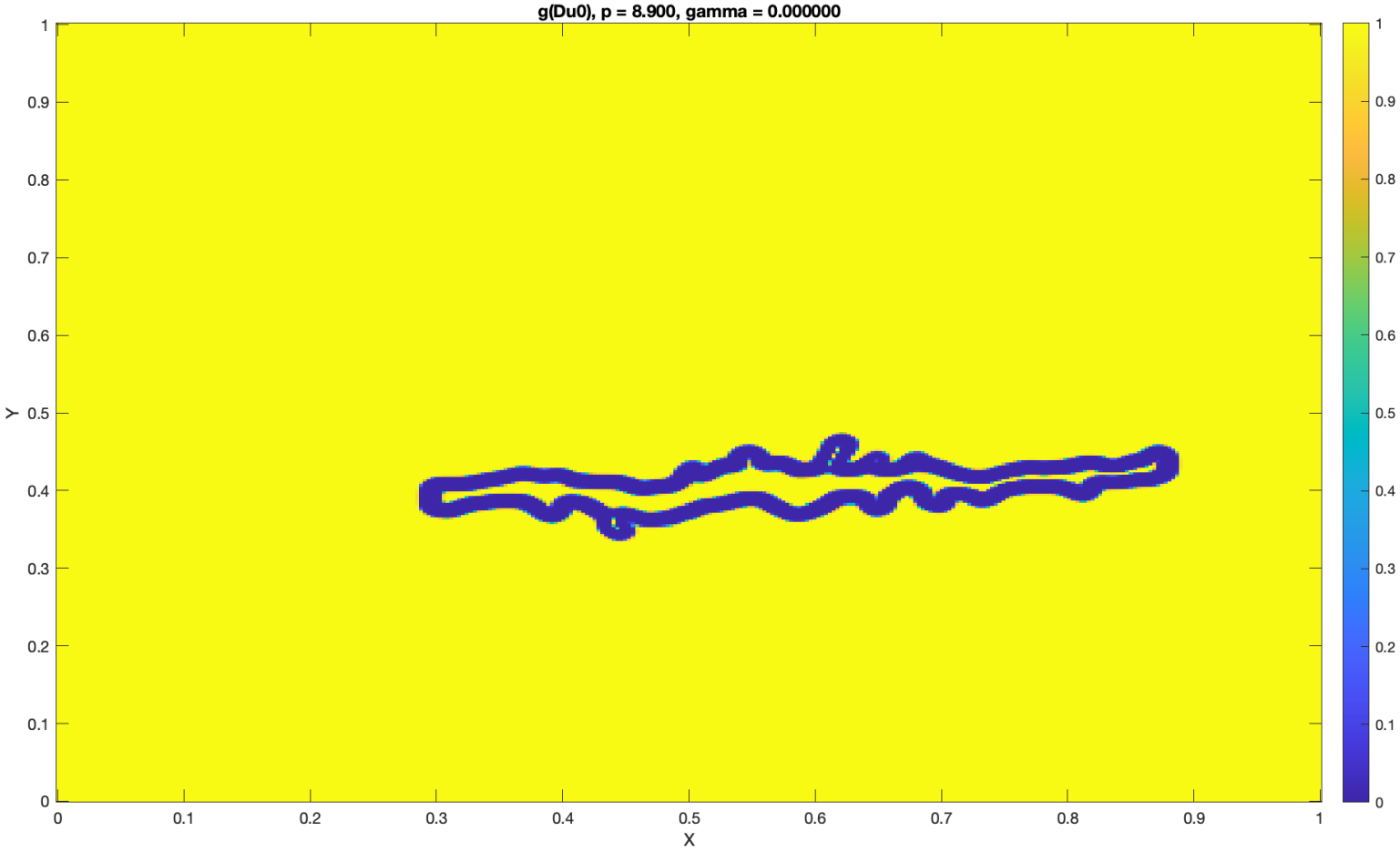
Run #4: Spine-Dendrite Image



Run #4: Spine-Dendrite Image



Run #4: Spine-Dendrite Image



Run #4: Spine-Dendrite Image



- $\sigma = 5.5 \times 10^{-5}$, $p = 8.9$, $\gamma = 5.5 \times 10^{-11.25}$

Summary

Challenges

- Choosing parameters for reasonable image segmentation p and γ
- Choosing the appropriate amount of smoothing without smoothing out important image features

To be completed...

- Implement the Mumford-Shah functional [1]

$$E[J, B] = C \int_D (I(\vec{x}) - J(\vec{x}))^2 \, \mathrm{d}\vec{x} + A \int_{D/B} \vec{\nabla} J(\vec{x}) \cdot \vec{\nabla} J(\vec{x}) \, \mathrm{d}\vec{x} + B \int_B ds$$

  optimizing this functional leads to a criteria for segmenting an image into sub-image regions and Ambrosio-Tortorelli give an algorithm for achieving the minimum and also show the minimum is well-defined.
- Implement on 3d geometries
- establish concrete relationship between γ, p, and d.
- Use convolution to blurr image instead of solving heat equation

References I

[1] L. Ambrosio and V. M. Tortorelli. Approximation of functional depending on jumps by elliptic functional via t-convergence. *Communications on Pure and Applied Mathematics*, 43(8):999–1036, 1990. doi: 10.1002/cpa.3160430805.

[2] J. Canny. A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(6):679–698, Nov 1986. ISSN 1939-3539. doi: 10.1109/TPAMI.1986.4767851.

[3] V. Caselles, R. Kimmel, and G. Sapiro. Geodesic active contours, 1997.

[4] Chenyang Xu and J. L. Prince. Gradient vector flow: a new external force for snakes. In *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pages 66–71, June 1997. doi: 10.1109/CVPR.1997.609299.

[5] L. Evans. *Partial Differential Equations*. Graduate studies in mathematics. American Mathematical Society, 2010. ISBN 9780821849743. URL https://books.google.com/books?id=Xnu0o_EJrCQC.

[6] R. A. Haddad and A. N. Akansu. A class of fast gaussian binomial filters for speech and image processing. *IEEE Transactions on Signal Processing*, 39(3):723–727, March 1991. ISSN 1941-0476. doi: 10.1109/78.80892.

[7] M. Kass, A. Witkin, and D. Terzopoulos. Snakes: Active contour models. *International Journal of Computer Vision*, 1(4): 321–331, 1988.

[8] S. Kichenassamy, A. Kumar, P. Olver, A. Tannenbaum, and A. Yezzi. Conformal curvature flows: From phase transitions to active vision. *Archive for Rational Mechanics and Analysis*, 134:275–301, 01 1996. doi: 10.1007/BF00379537.

[9] R. Klette. *Concise Computer Vision: An Introduction into Theory and Algorithms*. Undergraduate Topics in Computer Science. Springer London, 2014. ISBN 9781447163206. URL https://books.google.com/books?id=ZCu8BAAAQBAJ.

[10] R. LeVeque. *Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems*. Other Titles in Applied Mathematics. Society for Industrial and Applied Mathematics, 2007. ISBN 9780898716290. URL https://books.google.com/books?id=qsvmsXe8Ug4C.

[11] T. Morris. *Computer Vision and Image Processing*. Cornerstones of Computing. Macmillan Education UK, 2003. ISBN 9780333994511. URL https://books.google.com/books?id=MQGAQgAACAAJ.

[12] M. Nixon and A. Aguado. *Feature Extraction and Image Processing*. Electronics & Electrical. Newnes, 2002. ISBN 9780750650786. URL https://books.google.com/books?id=3JuesLhNXBAC.

References II

[13] S. Osher and R. Fedkiw. *Level Set Methods and Dynamic Implicit Surfaces*. Applied Mathematical Sciences. Springer New York, 2006. ISBN 9780387227467. URL https://books.google.com/books?id=i4bfBwAAQBAJ.

[14] L. Shapiro and G. Stockman. *Computer Vision*. Prentice Hall, 2001. ISBN 9780130307965. URL https://books.google.com/books?id=FftDAQAAIAAJ.

[15] G. Stockman and L. G. Shapiro. *Computer Vision*. Prentice Hall PTR, USA, 1st edition, 2001. ISBN 0130307963.

[16] H.-K. Zhao, T. Chan, B. Merriman, and S. Osher. A variational level set approach to multiphase motion. *Journal of Computational Physics*, 127(1), 8 1996. doi: 10.1006/jcph.1996.0167.